

Noah Mamie

✉ noah@mamié.ch

🌐 mamié.ch

in noah-mamie

🦋 @nmamie.bsky.social

Research Interests

Artificial Intelligence, Language Modeling, Recommender Systems, Knowledge Representation

Experience

- 2025 – Present 🔖 **Doctoral Researcher, University of Zurich.**
Focus: GenAI in the Context of Personalized News Recommender Systems.
- 2025 🔖 **GenAI Development Intern, Detecon Switzerland AG, Zurich.**
Focus: Retrieval Augmented Generation, Large Language Models, Intelligent Agents.
- 2023 – 2025 🔖 **Research Assistant, ETH-MTEC Chair of Applied Economics, Zurich.**
Focus: Knowledge Graph Retrieval, Recommendation Systems, LLMs, GNNs.
- 2021 – 2022 🔖 **Private Equity Intelligence DACH, Storskogen Switzerland AG, Zurich.**
Focus: Large-Scale Data Analytics, Digitalization, OPEX, Mergers & Acquisitions (M&A Research, Deal Flow, Due Diligence), and Optimization of Business Areas (Services, Trade, Industry).
- 2021 🔖 **Intern in Private Equity, Artum AG (Storskogen CH since 07/21), Zurich.**
Contributed to the merger of Artum and Storskogen (July 2021); Identified and contacted acquisition candidates; Analyzed target company data and trends; Developed company valuations and transaction structures.
- 2020 🔖 **Assurance Industrial Services Intern, Ernst & Young, Zurich.**
Conducted audits of annual and quarterly accounts per national and international standards (e.g., IFRS).
- 2019 🔖 **Student Consultant, Dr. Wolff Singapore Pte. Ltd., Singapore.**
Launched an e-commerce campaign for Plantur 21 (SEM, SEO, social media); Conducted primary research and grew the fan base; Presented weekly client updates.
- 2016 🔖 **Intern in Accounting, Marketing, and Sales, SpanSet, Villabona, Spain.**
Managed supplier bookkeeping and payment software; Coordinated with factory workers and sales team.

Education

- 2026 – Present 🔖 **PhD Excellence Program, Digital Society Initiative, University of Zurich.**
Selected for the competitive DSI Excellence Program, in recognition of outstanding students conducting interdisciplinary research.
- 2025 – Present 🔖 **PhD Candidate in Artificial Intelligence, University of Zurich.**
Supervisor: Prof. Dr. Abraham Bernstein. DDIS Group.
- 2022 – 2024 🔖 **Master of Science in Artificial Intelligence, University of Zurich.**
Final GPA: 5.54/6.0 - *Summa Cum Laude*
Thesis title: *The Society of HiveMind: Multi-Agent Optimization of Foundation Model Swarms to Unlock the Potential of Collective Intelligence.*
- 🔖 **Special Student in Computer Science, ETH Zurich.**
Relevant coursework: ML for Healthcare, Distinguished Seminar in Robotics.
- 2024 🔖 **Graduate Exchange Semester, University of Uppsala.**
Relevant coursework: Reinforcement Learning, Natural Comp. Methods for ML.

Education (continued)

- 2021 – 2022  **Computer Science, Additional Requirements, University of Zurich.**
Relevant coursework: Informatics (C and Python), Software Engineering (Java).
- 2017 – 2020  **Bachelor of Arts in Business Administration, University of St.Gallen.**
Thesis title: *Watermarking Deep Neural Networks*.
- 2019  **Undergraduate Exchange Semester, Singapore Management University.**
Relevant coursework: Finance, Industry consulting project.
- 2012 – 2016  **Bilingual Matura (EN-DE), Kantonsschule Enge, Zurich.**
Main Subject: Economics, Electives: Physics and Chemistry.

Research

Journal Articles

- 1 N. Mamié, P. Egger, and S. X. Rao, “The Winner Takes It All: Uncovering The Decision-Making Process Behind Nobel Laureate Choices Using Graph Reasoning,” Jun. 2024, Submitted to ACL-SRW 2024, In preparation for resubmission to PLOS One with additional experiment.  URL: <https://tinyurl.com/winner-takes-it-all>.

Conference Proceedings

- 1 N. Mamié and S. R. Xi, “The Society of HiveMind: Multi-agent Optimization of Foundation Model Swarms to Unlock the Potential of Collective Intelligence,” in *Advances in Swarm Intelligence*, Y. Tan and Y. Shi, Eds., Singapore: Springer Nature Singapore, 2025, pp. 243–254, ISBN: 978-981-95-0982-9.
- 2 E. Egli, N. Mamié, E. L. Dolev, and M. Müller, “Voting Booklet Bias: Stance Detection in Swiss Federal Communication,” in *Proceedings of the 8th edition of the Swiss Text Analytics Conference*, received best poster award, ACL Anthology, 2023.  URL: <https://arxiv.org/pdf/2306.08999>.

Workshops Organized

- 1 S. Rao, N. Mamié, Y. Tu, and P. Bhandar, *AI Support Systems for Academic Research*, 2024.  URL: <https://aclanthology.org/2024.swisstext-1.50.pdf>.

Invited Talks, Panels, Demonstrations and Public Engagement

- 2026  **Informfully Systems Demo**, Recommenders in News Media, Copenhagen (DK).

Reviewing

NeurIPS (2025), ESWC (2026), RecSys (2026), CIKM (2026).

Funding

- 2026  **50,000 CHF** UZH.ai Fellowship, UZH, Zurich.

Teaching Activities

- 2026 – Present  **Lecturer, Competence Center Human-Centered AI Systems, HSLU, Luzern.**
Guest Lecturer in the "Recommender Systems" course, presenting personalized recommender systems and demonstrating the Informfully ecosystem.
-  **Head TA, Department of Informatics, UZH, Zurich.**
Assisting Prof. Dr. Bernstein in organizing and teaching the first-semester course "Informatics and the Economy" (leading 8 Tutors; responsible for 1,400+ students).
- 2023  **Teaching Assistant, Department of Computer Linguistics UZH, Zurich.**
Assisted Dr. Mathias Müller for the course "Essentials of Text and Speech Processing".
- 2020  **Teaching Assistant, ITEM-HSG, St. Gallen.**
Assisted Prof. Dr. Wortmann in Fall 2020 (Skills: Operations Management, Excel).
- 2019  **Teaching Assistant, IWI-HSG, St. Gallen.**
Assisted Dr. Silic in Spring 2019 (Skills: Programming, Python, R, HTML/CSS/JS).

Featured Projects

- 2026  **Informfully: "Open source content delivery platform for reproducible science".**
Building a complete end-to-end ecosystem for reproducible experiments in recommender systems and content delivery research. *Python, Meteor.js, Vue.js, HTML.* 
- 2025  **HiveLLM: "The Society of HiveMind".**
Explored the foundations for a graph-based framework for building individual AI agents and orchestrating self-organizing, self-improving swarms. *Python.* 
- 2024  **Reinforcement Learning: "Playing Atari with Deep Reinforcement Learning".**
Trained a deep RL agent to successfully play the famous Atari game Pong. *Python.* 
-  **Vision Algorithms for Mobile Robotics: "A Visual Odometry Pipeline".**
Developed and demonstrated a complete visual odometry pipeline. *Python.* 
-  **ML for Healthcare: "Interpretable and Explainable Classification for Medical Data".**
Worked on a project for explaining deep learning models on medical data. *Python.* 
- 2023  **ZurichNLP: "Voting Booklet Bias".**
Developed a tool to analyze bias in voting booklets. *Python.* 
- 2022  **Chatbot: "Building an Intelligent Conversational Agent".**
Created an interactive chatbot with intelligent response capabilities. *Python.* 
-  **MyPark: "A Parking Reservation System".**
Built a web application for reserving parking spaces. *React.js, Java, Gradle, Heroku.* 
- 2020  **Bachelor's Thesis: "Watermarking Deep Neural Networks".**
Researched and implemented methods for watermarking deep neural networks. *Python.* 
- 2019  **Portfolio: "Data Visualization - Storytelling with R".**
Created a series of data visualizations focusing on effective storytelling. *R, Python Jupyter.* 

Skills

- | | |
|-----------------------|--|
| Languages |  German (native), English (bilingual), French (fluent), Spanish (fluent). |
| Programming Languages |  Python, C, Java, JavaScript, R, Matlab. |
| Databases |  MySQL, PostgreSQL, SPARQL. |
| Machine Learning |  PyTorch, PyTorch Geometric, TensorFlow. |
| Web Development |  HTML, css, React.js, TypeScript, JavaScript. |
| Other Tools |  L ^A T _E X, Gradle, Heroku, Google Cloud, SpringBoot, Linux OS. |

Volunteering

- 2022 – present  **Core Member, Global AI Hub.**
Active member of the 10million.AI initiative.
-  **Head of IT, Swiss Racketlon Federation.**
Board member responsible for IT, including website management, SEO, and data analytics.
- 2020 – 2021  **Head of Finance and Treasury, SGMUN Conference, St. Gallen.**
Managed budget and financial decision-making for the conference.
- 2018 – present  **Member of the Electoral Office, City of Wädenswil.**
Volunteered at polls and elections in Switzerland.
- 2015 – present  **Coach Diploma, Jugend+Sport Switzerland.**
Undertakes annual educational courses to enhance teaching methods for coaching badminton and table tennis to aspiring young players.

Miscellaneous Experience

Awards and Achievements

- 2026  **Fellowship Award**, Awarded a UZH.ai Fellowship in recognition of outstanding doctoral students at the University of Zurich whose research focuses on the development, application, or societal implications of Artificial Intelligence.
- 2023  **Best Poster**, Awarded as the "best poster" at the 8th Annual SwissText Analytics Conference.
-  **Datathon Winner**, 1st Place in the D-One Challenge of the ACE-ETH Datathon 2023.
- 2018  **Honorable Mention**, 2nd Place at St. Gallen Model United Nations Conference in the Committee on LAWS and Cybersecurity.

Certification

- 2020  **CS50x: Computer Science and the Art of Programming, HarvardX.**
Successfully completed MOOC covering introductory computer science concepts.